

# **Commerce XI Maths App**

## **Theory (Pages 24–26)**

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 36

### Question 1 - Let's Recall P24

A quantity whose value remains unchanged is called a

Options

- A. constant
- B. variable
- C. frequency
- D. deviation

**Correct answer: constant**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives constant.

### Question 2 - Let's Recall P24

A quantity that can take different values is called a

Options

- A. variable
- B. constant
- C. class boundary
- D. coefficient

**Correct answer: variable**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives variable.

### Question 3 - Let's Recall P24

The arithmetic mean is a measure of

Options

- A. central tendency
- B. dispersion only
- C. skewness only
- D. class width

**Correct answer: central tendency**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives central tendency.

## Question 4 - Let's Recall P24

Quartiles divide an ordered data set into

Options

- A. four equal parts
- B. two equal parts
- C. ten equal parts
- D. one hundred equal parts

**Correct answer: four equal parts**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives four equal parts.

## Question 5 - Let's Observe P24

An average alone cannot describe how observations are

Options

- A. scattered around it
- B. alphabetically arranged
- C. named
- D. rounded

**Correct answer: scattered around it**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives scattered around it.

## Question 6 - Theory P24

The amount by which observations deviate from an average is called

Options

- A. dispersion
- B. correlation
- C. classification
- D. tabulation

**Correct answer: dispersion**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives dispersion.

## Question 7 - Theory P24

Dispersion describes the degree to which numerical data tend to

Options

- A. spread about an average value
- B. equal the average
- C. form equal classes
- D. become constant

**Correct answer: spread about an average value**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives spread about an average value.

## Question 8 - Theory P24

Two series can have the same mean but

Options

- A. different dispersion
- B. the same observations
- C. different sample sizes only
- D. no variation

**Correct answer: different dispersion**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives different dispersion.

## Question 9 - Let's Construct P24

For comparing the scatter of two series, both series may first be plotted on a

Options

- A. number line
- B. pie chart only
- C. map
- D. calendar

**Correct answer: number line**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives number line.

## Question 10 - Theory P25

Which is an absolute measure of dispersion?

Options

- A. range
- B. coefficient of variation
- C. percentage
- D. ratio only

**Correct answer: range**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives range.

## Question 11 - Theory P25

The commonly used measures include range, quartile deviation, variance and

Options

- A. standard deviation
- B. median only
- C. mode only
- D. class mark

**Correct answer: standard deviation**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives standard deviation.

## Question 12 - Theory P25

Range is defined as

Options

- A. largest value – smallest value
- B. largest value + smallest value
- C. mean – median
- D.  $Q_3 - Q_1$  divided by 2

**Correct answer: largest value – smallest value**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives largest value – smallest value.

### Question 13 - Theory P25

If L is the largest value and S the smallest, range equals

Options

- A.  $L - S$
- B.  $L + S$
- C.  $S - L$
- D.  $LS$

**Correct answer:  $L - S$**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives  $L - S$ .

### Question 14 - Theory P25

Range is the

Options

- A. simplest measure of dispersion
- B. most detailed frequency table
- C. measure of central tendency
- D. same as variance

**Correct answer: simplest measure of dispersion**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives simplest measure of dispersion.

### Question 15 - Solved Example P25

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the smallest value is

Options

- A. 36
- B. 38
- C. 43
- D. 47

**Correct answer: 36**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 36.

### Question 16 - Solved Example P25

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the largest value is

Options

- A. 73
- B. 70
- C. 65
- D. 58

**Correct answer: 73**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 73.

### Question 17 - Solved Example P25

For weights 70, 62, 38, 55, 43, 73, 36, 58, 65, 47, the range is

Options

- A. 37
- B. 35
- C. 36
- D. 39

**Correct answer: 37**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 37.

### Question 18 - Solved Example P25

| Salary (₹00)     | 30-50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 |
|------------------|-------|-------|-------|--------|---------|---------|
| No. of employees | 7     | 15    | 30    | 24     | 18      | 11      |

For the grouped salary data, the lower limit used for the range is

Options

- A. 30
- B. 50
- C. 70
- D. 150

**Correct answer: 30**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 30.

## Question 19 - Solved Example P25

| Salary (₹00)     | 30-50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 |
|------------------|-------|-------|-------|--------|---------|---------|
| No. of employees | 7     | 15    | 30    | 24     | 18      | 11      |

For the grouped salary data, the upper limit used for the range is

Options

- A. 150
- B. 130
- C. 110
- D. 90

**Correct answer: 150**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 150.

## Question 20 - Solved Example P25

| Salary (₹00)     | 30-50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 |
|------------------|-------|-------|-------|--------|---------|---------|
| No. of employees | 7     | 15    | 30    | 24     | 18      | 11      |

The range of salaries is

Options

- A. 120
- B. 100
- C. 130
- D. 180

**Correct answer: 120**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 120.

## Question 21 - Theory P25

Range may be unsuitable when a data set contains

Options

- A. extreme values or outliers
- B. equal values
- C. a mean
- D. frequencies

**Correct answer: extreme values or outliers**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives extreme values or outliers.

## Question 22 - Theory P25

Quartile deviation is also called

Options

- A. semi-interquartile range
- B. total range
- C. mean deviation
- D. coefficient of variation

**Correct answer: semi-interquartile range**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives semi-interquartile range.

## Question 23 - Theory P25

Interquartile range equals

Options

- A.  $Q_3 - Q_1$
- B.  $Q_3 + Q_1$
- C.  $Q_2 - Q_1$
- D.  $(Q_3 - Q_1)/(2)$

**Correct answer:  $Q_3 - Q_1$**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives  $Q_3 - Q_1$ .

## Question 24 - Theory P25

Quartile deviation equals

Options

- A.  $(Q_3 - Q_1)/(2)$
- B.  $Q_3 - Q_1$
- C.  $(Q_3 + Q_1)/(2)$
- D.  $(Q_2)/(2)$

**Correct answer:  $(Q_3 - Q_1)/(2)$**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives  $(Q_3 - Q_1)/(2)$ .

## Question 25 - Theory P25

Quartile deviation reduces the influence of

Options

- A. extreme values
- B. the central 50% values
- C. class widths
- D. all frequencies

**Correct answer: extreme values**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives extreme values.

## Question 26 - Solved Example P25-26

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91,  $Q_1$  is

Options

- A. 58
- B. 56
- C. 60
- D. 61

**Correct answer: 58**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 58.

## Question 27 - Solved Example P25-26

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91,  $Q_3$  is

Options

- A. 79
- B. 72
- C. 75
- D. 84

**Correct answer: 79**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 79.

## Question 28 - Solved Example P25-26

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, the interquartile range is

Options

- A. 21
- B. 10.5
- C. 19
- D. 27

**Correct answer: 21**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 21.

## Question 29 - Solved Example P25-26

For the ordered marks 52, 56, 58, 60, 61, 67, 72, 75, 79, 84, 91, the quartile deviation is

Options

- A. 10.5
- B. 21
- C. 9.5
- D. 11

**Correct answer: 10.5**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 10.5.

## Question 30 - Theory P26

For an open-end frequency distribution, an appropriate measure of dispersion is

Options

- A. quartile deviation
- B. range
- C. largest value
- D. smallest value

**Correct answer: quartile deviation**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives quartile deviation.

### Question 31 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

[[line:The total frequency N is]]

Options

- A. 1000
- B. 948
- C. 816
- D. 1052

**Correct answer: 1000**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 1000.

### Question 32 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

The first quartile position is

Options

- A. 250th observation
- B. 125th observation
- C. 500th observation
- D. 750th observation

**Correct answer: 250th observation**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 250th observation.

### Question 33 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

The first quartile class is

Options

A. 70-90

B. 50-70

C. 90-110

D. 110-130

**Correct answer: 70-90**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 70-90.

### Question 34 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

The third quartile class is

Options

A. 110-130

B. 90-110

C. 130-150

D. 70-90

**Correct answer: 110-130**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 110-130.

### Question 35 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

$Q_1$  is approximately

Options

A. 84.24

B. 70

C. 89

D. 110

**Correct answer: 84.24**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 84.24.

### Question 36 - Solved Example P26

| Income (₹)     | Less than 50 | 50-70 | 70-90 | 90-110 | 110-130 | 130-150 | Above 150 |
|----------------|--------------|-------|-------|--------|---------|---------|-----------|
| No. of persons | 59           | 102   | 125   | 330    | 200     | 132     | 52        |

$Q_3$  is approximately

Options

A. 123.4

B. 110

C. 130

D. 84.24

**Correct answer: 123.4**

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 123.4.